MIL-E-1/507F(EL)
9 January 1970
SUPERSEDING
MIL-E-1/507E(SigC)
26 March 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON, PULSE
TYPE 7452A

The complete requirements for procuring the electron tube described herein shalll consist of this document and the latest issue of Specification MIL-E-1.

Description: 16.0 GHz fixed frequency, 85 kW peak power output, integral 2-symmetrical-lobe magnet, air cooled.

Outline and dimensions: See Figure 1 herein. Weight: 25 lbs., approx.

Absolute Ratings: (Note 1)

Parameter:	Ef	eb	if(surge)	tk	Anode T	VSWR	Cathode (bushing)T
Unit:	V	kv	a	sec	°C	ratio	°C
Maximum:	5.5	30	24	---	+125	1.5:1	+200
Minimum:	---		---	180	---	---	---
	(Note 2, 3)		(Note 3)		(Note 4)		(Note 4)

Parameter:	Ib	ib	Pi	pi	Du	tpc	Pressuriz. (Output)
Unit:	mAdc	a	W	kw	---	usec	psia
Maximum:	30	14	780	315	0.0028	0.6	45
Minimum:	---	10	---	---	---	---	14.7
						(Note 5)	(Notes 6, 7)

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TEST CONDITIONS: (Notes 8, 9)

Parameter:	Ef	Ib	VSWR	Du	tpc	tfc	trv	F
Unit:	V	mAdc	ratio	---	usec	usec	usec	GHz
Maximum:	4.8	26	1.1:1	---	0.275	0.15	0.14	16.160
Minimum:	---	22	---	0.00215	0.225	---	0.09	15.840
	(Note 2)	(Note 2)	(Notes 7, 10)					

Mounting Position: Vertical (Note 9) Input Connection: (See Fig. 1 herein)

Output Connection: Waveguide choke coupling flange, modified UG-541/U (See Fig. 1 herein)

Cooling: Forced-air (See Fig. 1, Note 14 herein) Magnet Isolation: (Note 11)

Mounting Provision: (See Fig. 1, Note 20, herein). Handling-Stress Caution: (Note 12)

Method	Requirement or test	Conditions	AQL % defective	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
	GENERAL					---	---	---
---	Preproduction sample approval	Required;(Note 13)	---	---	---			
---	Marking	(Notes 14, 15)	---	---	---	---	---	---
---	Dimensions	(See Fig. 1)	---	---	---	---	---	---
---	Packaging	(Note 16)	---	---	---	---	---	---
	QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 17)							
---	Visual-mechanical examination	See Fig. 1	---	---	---	---	---	---
4003	Pressurizing	Air; 50 psia, min. Output assembly (Note 18)	0.65	II	---	---	---	---
4289	Heater current	Ef=4.8V tk=180 sec., min. (Note 3)	0.65	II	I_f	10.8	13.2	A

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QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 17) (Cont'd)

Method	Requirement or test	Conditions	AQL % defective	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
---	Oscillation (1):	VSWR=1.1:1 max. (Notes 1, 2, 8)	0.65	II	---	---	---	---
4303	Heater-cathode warmup time	Ef=4.8V tk=180 sec., max. Ef=0.0V for test	---	---	---	---	---	---
---	Frequency	Anode T=+70°±15°C (Notes 4, 19)	---	---	F	15.840	16.160	GHz
4304	Pulse characteristics	(Notes 20, 21)	---	---	---	---	---	---
4306	Pulse voltage	---	---	---	epy	22	26	kv
---	Average anode current	Ib=26 mAdc	---	---	---	---	---	---
4250	Average power output	---	---	---	Po	150	---	W
4308	Spectrum measurement: RF bandwidth	VSWR=1.5:1 (Note 22)	---	---	BW	---	8	MHz
4315	Stability	VSWR=1.5:1 (Note 23)	---	---	MP	---	0.75	%

QUALITY CONFORMANCE INSPECTION, PART 2

1031	High-frequency vibration:	No voltages (Note 24)	6.5	S3	---	---	---	---
4308	RF bandwidth	VSWR=1.5 Ib=22-26 mAdc			BW	---	8	MHz

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QUALITY CONFORMANCE INSPECTION, PART 2, (Cont'd)

Method	Requirement or test	Conditions	AQL % defective	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
4310	Frequency pulling factor	(Osc.(1)) VSWR=1.5:1 (Note 25)	6.5	S3	ΔF	---	25	MHz
4311	Frequency pushing factor	(Osc.(1)) VSWR=1.1:1, max. Swept: 22 to 26 mAdc at 50 Hz rate	6.5	S3	ΔF	---	3.5	MHz/mA

QUALITY CONFORMANCE INSPECTION, PART 3, LIFE

---	Cycling life	Group D (Osc.(1)) VSWR=1.5:1 min. cycled thru $\frac{\lambda g}{2}$ ea. 15 minutes of Oscillation (Note 26)	---	---	N	175	---	cycles
(4.6.2)	Life-test End-points (Osc.(1)):	Aver. Power Output			Po	120	---	W
		Pulse voltage			epy	22	26	kv
		Fixed tuned frequency			F	15.840	16.160	GHz
		RF bandwidth			BW	---	8	MHz
		Stability			MP	---	1.0	%
		Freq. Drift, (Note 27)			ΔF	---	2	MHz/hr

QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC CHECK, (NOTES 28, 29, 30)

1006	Salt spray (corrosion)	---	---	---	---	---	---	---
1011	Humidity	---	---	---	---	---	---	---
1042	Shock	Normal mtg. Test Cond.A except G=15(peak) (Note 31)	---	---	---	---	---	---
4266	Input capacitance	Heater to anode	---	---	C	10	20	pF

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Method	Requirement or test	Conditions	AQL % (defective)	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC CHECK, (Notes 28, 29, 30)Contd.								
4027	Temperature coefficient	(Osc.(1)) VSWR=1.1:1 Anode T=+55° to +85°C (Note 4)	---	---	Δ F/°C	---	0.75	MHz/°C
---	Low pressure	(Osc.(1)) VSWR=1.5:1 Pressure=522 mmHg (Note 32)	---	---	---	---	---	---
1026	Low-temperature operation	Pre-test cooling = 60 minutes at -65°C (Osc.(1)); Stability test reqts; tk=180 sec., max. (Note 23)	---	---	MP	---	2.0	%
---	Frequency	(Osc.(1)) (Notes 4, 19, 33)	---	---	F	15.840	16.160	GHz
---	Air cooling	(Osc.(1)) T_A=+25°C Pi=624W±10% Air flow=40 cfm (Note 34)	---	---	ΔT	---	+70	°C

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NOTES:

1. Refer to Specification MIL-E-1, Section 6 "Note" covering Absolute Ratings.

2. Prior to application of high voltage, the filament current applied should be between 10.8 and 13.2 amperes. The procedure should include, first, applying 4.8 volts for 180 seconds minimum to filament; then, upon application of anode voltage, the heater voltage should be reduced until the anode current is 26 mAdc.

3. the heater surge current should not be allowed to exceed 24 rms amperes.

4. Refer to Figure 1, Notes 12, 21, herein for temperature-measurement point. Temperature shall be measured after thermal equilibrium has been reached.

5. The characteristics of the applied pulse must be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of the pulse voltage fall are among the more important considerations. Negative post-pulse voltage oscillations may cause leakage current and noise. The established pulse characteristics (refer to Oscillation (1) pulse characteristics, herein) should be adhered to.

6. The gas used in pressurization shall provide insulating properties equal to or better than that of dry air at the pressure specified.

7. The waveguide must be pressurized at 14.7 psia, minimum, to prevent breakdown under matched conditions, VSWR=1.1 maximum. Under mismatch conditions, VSWR=1.5 maximum, the waveguide must be pressurized at 35 psia, minimum.

8. Unless otherwise specified for a particular test parameter (see testing program table herein), the cooling and pressurization requirements-data indicated herein shall be effective wherever applicable throughout the testing program.

9. The tube shall be mounted with the cathode vertical to within 15° deviation.

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Notes (contd)

10. Output window shall be pressurized at 14.7 psia.

11. During handling, transportation, mounting, and use, care must be exercised to prevent demagnetization. Ferromagnetic materials should not be brought within 8 inches of the tube magnet, except at the ends where the clearance may be 3 inches. Energized magnets must not at any time be permitted to be closer than 12 inches from the tube magnet.

12. Care should be exercised during all handling for installation, storage, and use of the tube, to prevent greater shock or vibration than that of the values for which the tube is designed.

13. Preproduction Sample Approval in lieu of Qualification.- Approval of preproduction sample, based upon preproduction sample inspection, shall replace Qualification requirements of Specification MIL-E-1. Preproduction Sample tests, testing, sampling, and approval criteria shall correspond directly with all pertinent requirements in the specification as normally applicable to qualification of product.

a. Preproduction tests will be performed after award of contract, at the manufacturer's plant, unless otherwise specified in the bid request or contract. Approval of the preproduction sample is required prior to initial delivery of units on the contract. The fabrication of production units, prior to approval of the preproduction sample, shall be at the manufacturer's risk. Unless otherwise specified in the bid request or contract, the specimens shall become the property of the government and, upon approval, shall serve as a reference standard for the contracting officer and government inspector.

b. Requests for information regarding preproduction testing and approval of the product(s) covered herein should be referred to the Commanding General, US Army Electronics Command, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, ATTN: AMSEL-PP-EM-1.

c. No Qualified Products List (QPL) entry will be undertaken for the product(s) covered herein as approval of design and performance of product will be based upon Preproduction Sample Approval.

d. In no manner shall the above requirements for Preproduction Sample Approval be construed as affecting requirements for production acceptance inspection of the product(s) covered herein.

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NOTES (Contd)

14. Tube-designation marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

15. Tube identification and tube serial number shall be marked in the most practical and most readily discernible location on the tube, relative to its normal mounted position (see Note 9, above).

16. Unless otherwise specified in contract or order, each tube shall, as a minimum protection against moisture and moisture vapor be sealed within a moisture-vaporproof barrier bag, or equivalent, normally acceptable under packaging requirements of Specification MIL-E-75. Also each electron tube shall be furnished with a dust cover for protection of the output connector.

17. The AQL for the combined detectives for attributes in Quality Conformance Inspection, Part 1, exclusive of "Visual-Mechanical examination", shall be 1.0%. A tube having more than one defect shall be counted as one defective.

18. The tube shall be gasketed, and test made of the output assembly. The specified pressure shall be applied, and the tube immersed in water. The tube shall be rejected if evidence of continuous formation of air bubbles exists.

19. The frequency shall be within the limits specified when measured under the stipulated temperature condition(s) and in accordance with paragraph 4.9(h) of Standard MIL-STD-1311.

20. This test need be conducted under only one set of conditions within the limits presented for Oscillation(1) herein.

a. The characteristics of the applied pulse must be those that result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. Negative post pulse voltage oscillation should be minimized to avoid leakage current and noise. Also, values of voltage and current fall time, spike, ripple, inverse voltage and post voltage oscillations should be minimized.

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NOTES (Contd)

b. In instances where the pulse shape departs appreciably from the rectangular, an equivalent rectangular pulse with minimum departure areas shall be used for determination of pulse width and peak pulse current.

c. Any capacitance used in the viewing system shall not exceed 6.0 pF.

21. The pulse characteristics for Oscillation (1) test shall be as follows:

	Min	Max	Unit
tpc	0.225	0.275	usec
tfc	---	0.15	usec
trv	0.09	0.14	usec
Ripple	---	±10	%
Inverse voltage	---	±20	%
Post pulse forward voltage	---	±5	%

22. The r-f bandwidth shall be within the limits specified when the specified VSWR is introduced in the load at a distance of approximately 1/4-meter from the magnetron, with the load phase adjusted for maximum spectrum degradation.

23. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. Output (r.f.) pulses are considered to be "missing" if, due to any causes, the average r.f. pulse energy, within a ±1 percent frequency range of the normal operating frequency, is less than 70 percent of the energy level of a normal pulse. The percentage of missing pulses shall not exceed the amount specified, when measured in the last 3 minutes of a test interval not to exceed 6 minutes. Observation shall be made at the load phase position corresponding to maximum instability.

24. After subjection to Vibration test, each tube shall meet requirements for r-f spectrum and stability per Oscillation (1) testing requirements herein.

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25. The pulling measurement test shall be made in a manner such that thermal effects do not contribute appreciable error.

26. One Life Test cycle shall consist of operation under the conditions of Osc (1) testing, as follows:

Ef Volts	Ib mAdc	Period (Minutes)
4.8	0	3
0	26	120
0	0	27

a. The input power to the heater shall be snapped on at the beginning of each Life Test cycle with a surge current of 24 to 26 amperes RMS.

b. The standing wave introducer shall be moved during the test so that operation is obtained for load phase positions corresponding to maximum power output, minimum power output, maximum frequency, and minimum frequency. If automatically driven, the standing wave introducer shall be cycled continuously through a line length approximately one-half wave length long at a maximum rate of four cycles per hour; if manually moved, the standing wave introducer shall be cycled through the four load phase positions corresponding to maximum power, minimum power, maximum frequency, and minimum frequency at least once during the specified life of the tube, spending approximately equal periods of time in each phase position.

27. This test shall be performed in such a manner as to exclude thermal effects. The frequency drift shall not exceed the amount specified when averaged over each eight hour period during the minimum specified life, except during the first 24 hours. The frequency drift during the first 24 hours shall not exceed 6.0 MHz per hour when averaged over each eight hour period.

28. For periodic tests, sample tubes are to be selected at random intervals so that during only twelve-month period a minimum of two tubes shall have been successfully tested. In the event of failure of one or more tubes in a sample, these tests are to become a part of Quality Conformance Inspection, part 2, until two successive lots have been successfully completed on these tests. After two successive lots have thus been successfully completed, the periodic tests shall then be eligible to be resumed on a twelve-month basis.

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29. After subjection to all Periodic-check tests, the sample tube(s) shall meet the requirements of all tests in QCI, Part 1 herein.

30. Although all tests in Part 3, Periodic check, herein may be performed in any sequence, the sequence recommended is as follows:

Air cooling
Input capacitance
Frequency
Temperature coefficient
Low pressure
Low-temperature operation
Shock
Salt spray (corrosion)
Humidity

However, at option of the manufacturer, the tube sample (s) may be subjected to the Part 3, Periodic Check tests herein in accordance with the following grouping plan (see c, below):

a. The tube sample(s) shall be subjected first to all tests in Group 1, then to all tests in Group 2, and in the test sequence shown within each group. The requirement of Note 29 above shall be effective for performance immediately after the last test in each group.

b. Group 1

Air cooling
Input capacitance
Low pressure
Frequency
Temperature coefficient

Group 2

Low-temperature operation
Shock
Salt-spray (corrosion)
Humidity

c. The above grouping plan is provided to relieve the need to repeat pre- and post-test end-point measurements (all tests of QCI-1 herein) normally pertinent to individual tests such as Shock and Humidity.

31. The magnetron shall be mounted on a test plate and subjected to five half-sine shock pulses each in each of three mutually perpendicular axes parallel to the respective reference planes shown in Fig. 1 herein.

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32. Under specified pressurization and with load phase varied through 1/2 λ g, (waveguide), minimum, there shall be no evidence of breakdown in input assembly.

33. The output frequency of the electron tube shall be within the specified limits when the anode-block temperature is varied from +55° to +85°C.

34. Δ T is the temperature rise of the anode above that of the coolant temperature at inlet of the cooling fins with the airflow as specified.

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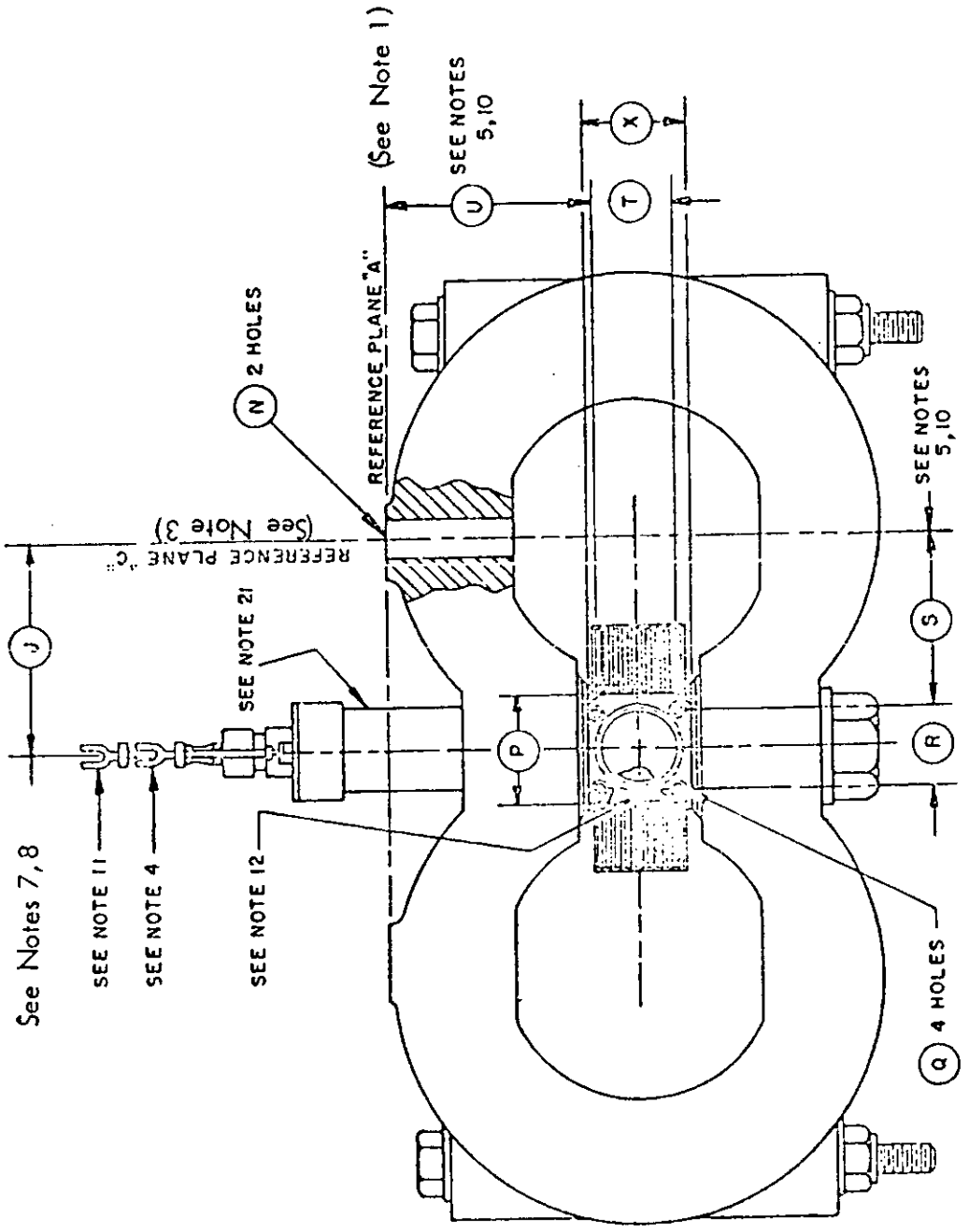


Figure 1. Outline and dimensions.-(Front view).

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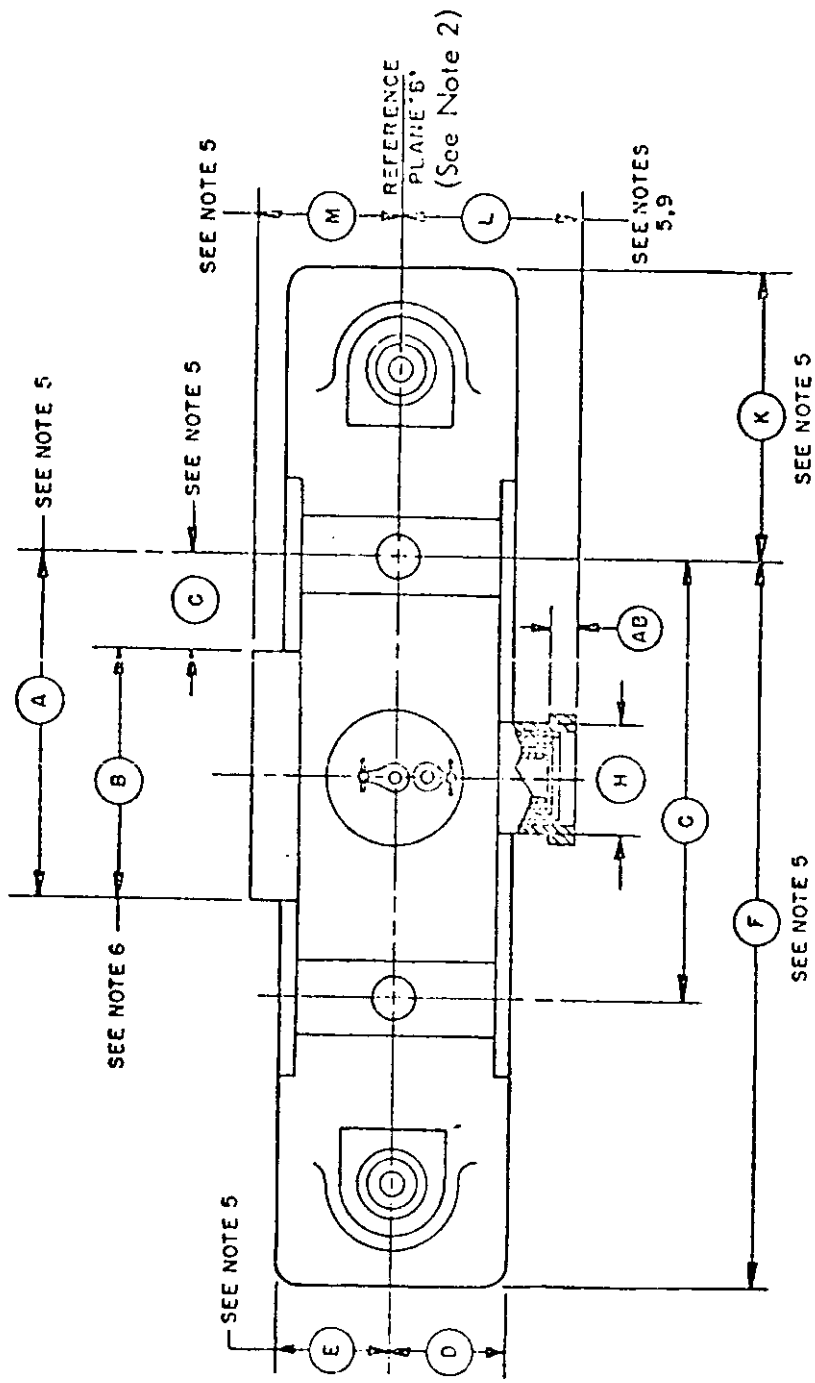


Figure 1-(cont'd). Outline and dimensions.-(Top view)

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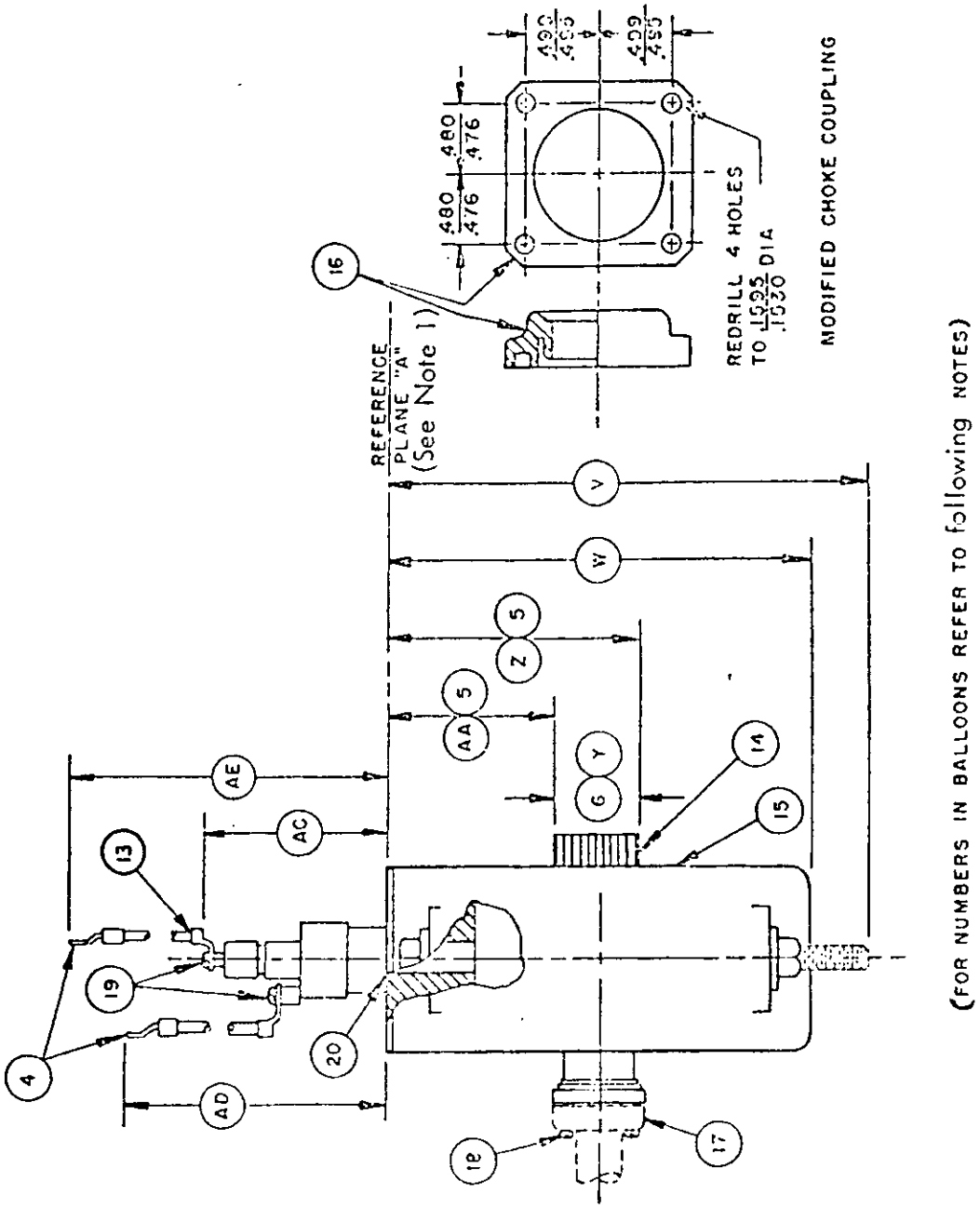


Figure 1-(Cont'd) Outline and dimensions.-(Sideview)

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LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECTION PART 1		
F		8.875 (225.425)
K		3.625 (92.075
L	1.870 (47.498)	1.970 (50.038)
Q	6-32 NC-28	
R	.946 (24.028)	.966 (24.536)
S	2.077 (52.755)	2.217 (56.311)
T	.984 (24.993)	1.004 (25.501)
U	2.499 (63.474)	2.639 (67.030)
AC		3.312 (84.124)
AD	7.750 (196.850)	8.250 (209.550)
AE	7.250 (184.150)	7.750 (196.850)
QUALITY CONFORMANCE INSPECTION PART 3, (PERIODIC CHECK)		
H		1.100 (27.940)
P	1.297 (32.943)	1.327 (33.705)
AB	.173 (4.394)	.203 (5.156)
X	1.297 (32.943)	1.327 (33.705)

LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECTION PART 2		
A		4.281 (108.737)
B	2.930 (74.422)	3.070 (77.978)
C	.963 (24.460)	
D		1.625 (41.275)
E		1.625 (41.275)
J	2.625 NOM	
M		2.000 (50.800)
N	.469 (11.912)	.531 (13.487)
V		7.250 (184.150)
W		6.500 (165.100)
Y	.985 (25.019)	1.135 (28.829)
Z		3.780 (96.012)
AA	2.345 (59.563)	
G	5.178 (131.521)	5.322 (135.178)

FIG. 1-(contd.). LEGEND FOR CODED DIMENSIONS.

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Fig. 1 NOTES:

1. Reference plane "A" is defined as a plane passing along the finished surface of the magnet.

2. Reference plane "B" is defined as a plane passing through the centers of the mounting holes, and perpendicular to reference plane "A".

3. Reference plane "C" is defined as a plane passing through the center of the mounting hole, and mutually perpendicular to reference planes "A" and "B".

4. Wire: No. 11 AWG, 7 x 6 x 30 AWG or equal.
   Insulation: NEMA 10, white silicone rubber, fiberglass tubing grade 4A1 or equal.
   Lugs: Burndy spade lug YAV10-T21F or equal; one lug painted brown to identify it as common heater-cathode connection, outer terminal.

5. Tolerance includes angular as well as lateral deviation(s).

6. These dimensions apply to radiator size only.

7. This dimension refers to centerline of tube.

8. Parts on this centerline may vary (displacement) by 1/8 in.

9. Surface of output flange to be parallel to reference plane "B" within 0.020-in.

10. Tube-to-tube variations shall be within the dimensional tolerances, but hole-to-hole variation on any single tube shall not vary more than 0.030-in. within the dimensional tolerances.

11. Heater connection.

12. Point at which temperature shall be measured on anode body.

13. Minimum distance from high-voltage to ground should be 1.500-in through air, at normal atmospheric pressure.

14. Anode cooling-fin inlet-duct for supplying air to the tube cooling system must be such as to insure delivery of air to the entire cross-section of the inlet, with minimum side leakage.

15. Anode must be grounded effectively, such as through the mounting holes.

Fig. 1-(cont'd). Outline and dimensions.

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Fig. 1 Notes-(Cont'd)

16. Waveguide choke coupling flange, -modified UG-541/U.

17. All joints of waveguide output assembly to be hermetically tight. "O"-ring silicone-rubber gasket, per Specification Sheet MIL-F-3922/59 shall be used for test.

18. Four #6-32 screws required to attach waveguide choke to tube.

19. High-voltage terminals: center terminal=heater, outer terminal=heater-cathode.

20. Mounting holes: Entire support and fastening of tube to be effective therein.

21. Point at which temperature shall be measured on cathode bushing.

Fig. 1-(Cont'd). Outline and dimensions.

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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-1/507F(EL), ELECTRON TUBE, MAGNETRON, PULSE TYPE 7452A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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